

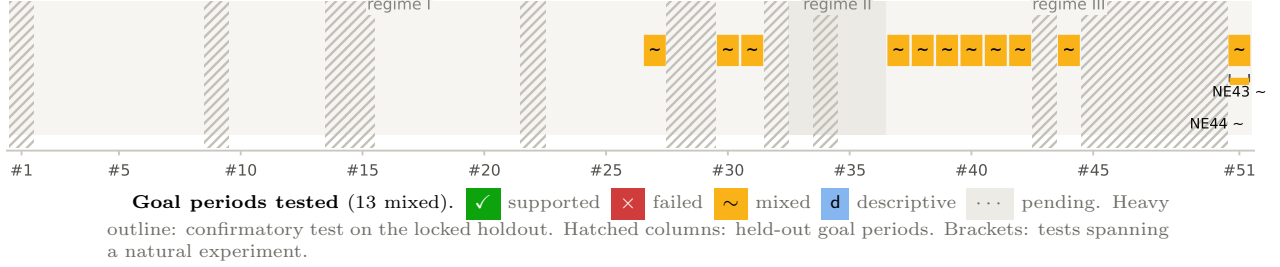
H16: Agents get stuck in metastable traps and escape by Kramers kinetics; nudges lower the barrier

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 48% Faithfulness 1.5/5 Usefulness 2.5/5

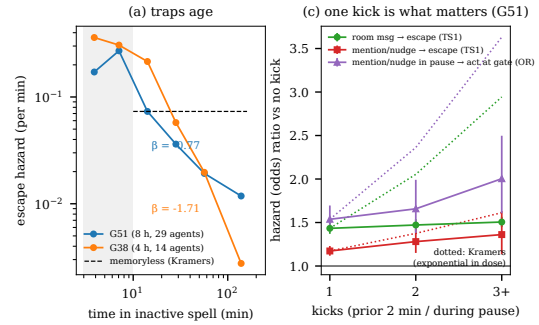
Bottom line. Traps age by themselves, not by Kramers escape; one directed kick at a pause gate helps, and real failure loops age only in #51.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

The units are per-agent trap spells: inactive spells, timer-pause chains, error loops and repeated-command loops. We test them in 11 non-holdout goal periods. Kramers escape predicts a constant escape hazard. Instead, traps *age*: within an agent, the hazard falls as about $t^{-0.8}$ (#51 $\beta = -0.77$, #38 -1.71). Each re-pause lowers the odds of action at the next timer expiry (#51 -0.38). One directed message during a pause raises those odds $\times 1.5$ – 2.9 (5/7 periods); more messages add little. Round 1b counts real failures instead of stderr, takes the leading-@ nudge target and adds Jev blocked and loop spells. Error-loop aging was mostly a stderr artifact: loop rows fall 51%, and aging survives only in #51 (-0.57). Jev blocked spells age in #51 and #27. The kick results do not change. With the nudger off (NE43), aging does not change. Mean-field $\beta J_0 = 0.09$ – 0.41 gives no bistability. The holdout is not run.



Observables. (a) Escape hazard from inactive spells vs time in the spell, in the two periods with the most power. A spell ignores single stray actions. After the first minutes (grey), Kramers predicts the flat dashed line. The measured hazard falls, so traps age. (b) #51 hazard and odds ratios vs the number of kicks. Dotted lines extrapolate Kramers (exponential in dose) from one kick. The measured curves flatten: the first message carries the effect.

PHYSICAL MODEL

Models 01/02 (mean field) with Kramers escape. The activity x of an agent moves in a double well $U(x)$ (idle and active basins) at temperature T . A trap is a stay in the idle basin. Escape has rate k over barrier ΔU , with attempt rate ν . A kick (a message that reaches the agent) adds a force f . Swarm: Curie–Weiss coupling J_0 , field h , active fraction m . Fit: hazard h_{it} , agent effects α_i , aging slope β , dose effects γ_n .

$$k = \nu e^{-\Delta U/T}, \quad G(x) = -\ln P(x) = U(x)/T + \text{const}$$

$$k(n) = k_0 e^{nf/T} \text{ (Kramers)} \quad \text{vs} \quad k(n) = k_0(1 + cn) \text{ (triggers)}$$

$$\ln[-\ln(1 - h_{it})] = \alpha_i + \beta \ln t + \sum_n \gamma_n \mathbb{K}[\text{dose}_{it} = n]$$

$$f(m) = -\frac{1}{2}J_0 m^2 - hm - Ts(m); \text{ bistable iff } \beta J_0(1 - m^2) > 1$$

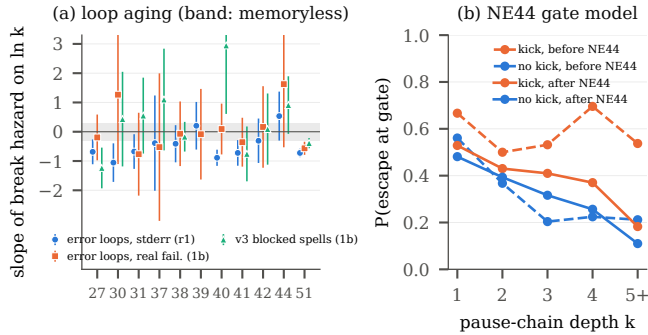
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

Kick a stuck agent early: its escape hazard falls as about $t^{-0.8}$ with time in the trap. Send one directed message during a pause. It raises the odds of action at the next timer expiry $\times 1.5$ – 2.9 ; repeats and room messages add little. Under the 5-min pause default (after NE44), a kick lifts escape from deep pause chains only from 0.17 to 0.25. Messages do not break real failure loops (9/9 periods), so use tooling or a reset there. The aging is the agent's own: it does not change with the nudger off. The physics adds little, because Kramers fails and aging with saturating triggers is ordinary survival analysis.

H16-metastable-traps-kramers · generated 2026-10-04 12:03 by infra/summaries/build_summaries.py · card hypotheses/H16-metastable-traps-kramers/README.md · ratings: H16 re-evaluation agent RE-B1 (suggested; round-1 values were coordinator-calibrated)

RESULTS VS PREDICTIONS

prediction (round 1b)	observed	verdict
real-failure loops (majority)	age 3/10 point, 0/10 CI; #51 failed -0.57	
messages do not break them	9/9 periods	holds
Jev blocked spells (#51, #38, $\geq$ half)	age #51 -0.39, #27 -1.24; #38 failed flat; timer-like in 3	
Jev loop spells (#51)	age -0.46 [-0.55, -0.18]	partial
gate kick OR (leading @)	≥ 1.5 5/7 (1.5-2.9)	holds
room kicks above null	6/8	holds
NE44 kick $\times \ln k$: then < 0	0 +0.57 before, +0.06 after	failed
NE44 declared pauses shrink	90 s $\rightarrow$ 180 s	failed
NE43 aging without TS1r kicks	-0.59 $\rightarrow$ -0.53; gates -0.31 $\rightarrow$ -0.26	holds
NE43 gate escape drops	+0.006 [-0.09, 0.10]	failed



Second figure. (a) Loop aging slope per period: error loops on stderr (round 1), real failures (round 1b) and Jev blocked spells. The gray band is memoryless. Only #51 keeps a clear negative slope on real failures. (b) Gate escape vs pause-chain depth. Before NE44, a directed kick rescues agents at any depth. After NE44, deep chains barely respond.

GOAL PERIODS IN DETAIL

G27 $\sim$ 1b: mixed (unchanged under the round-1 rule; its core...
G30 $\sim$ 1b: mixed (unchanged under the round-1 rule; its core...
G31 $\sim$ 1b: mixed (unchanged under the round-1 rule; its core...
G37 $\sim$ 1b: mixed (unchanged under the round-1 rule; its core...
G38 $\sim$ 1b: mixed (unchanged under the round-1 rule; its core...
G39 $\sim$ 1b: mixed (unchanged under the round-1 rule; its core...
G40 $\sim$ 1b: mixed (unchanged under the round-1 rule; its core...
G41 $\sim$ 1b: mixed (unchanged under the round-1 rule; its core...
G42 $\sim$ 1b: mixed (unchanged under the round-1 rule; its core...
G44 $\sim$ 1b: mixed (unchanged under the round-1 rule; its core...

+3 more periods: see the card.

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	1
E interventional	1		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (`writup/paper.tex`).

CAVEATS

The NE44 contrast compares periods, because the change sits inside the locked holdout window. Day length (4 h vs 8 h), roster and goals confound it. Jev `p.blocked` means “waiting on others” in regime I and “failing or looping” in regime III. Window spells are short in 4-h periods. Kick labels are temporal coincidences. Several 3–5-day periods have unstable day-bootstrap CIs; we report Wald CIs alongside. We did not re-run the barrier-inversion and swarm blocks, because their inputs did not change.

ROUND 2

Test self-reinforcement with a Pólya urn on the context ledger: own-output share vs the aging slope. Test content-level spirals on the holdout (#45).

- **Essence.** Traps deepen by self-reinforcement as the context fills with the agent’s own output.
- **H16-R1** Pólya urn: $P(\text{repeat}) \propto$ the share of the context that is own repeats.
- **H16-R2** One directed message works because it dilutes the urn; repeated messages add little once the context is diluted.
- **H16-R3** Trap depth is a model trait: some models loop more.